Coffee Sales Research Report

STQD6414 Data Mining

Members:

Liang Haizhu	P156872
Dong Qianqi	P161798
Zhang Ruixuan	P165398
Mhamad Shhab Aldeen Hasan	P166175

目录

1. Introduction

1.1. Research Background

The coffee market is fiercely competitive today. Focusing solely on sales figures makes it difficult to make truly precise business decisions. As consumers increasingly diversify their choices, we must understand how people transition from “casual visitors” to “loyal customers”—a key factor in boosting repeat purchase rates.

This study directly utilizes real transaction data from a specific coffee brand. By uncovering temporal patterns within orders, we aim to clarify how members' consumption habits evolve over time.

1.2. Research Questions

1.2.1. Based on real transaction data of a coffee brand, explore the temporal evolution patterns of coffee category choices and the core preference transition paths as members move from their first purchase to long-term consumption;

1.2.2. identify key coffee categories and consumption stage characteristics that influence members' repeat purchase behavior;

1.2.3. build a sales forecasting model suitable for this coffee brand (using latte as an example) to predict future short-term sales trends and provide data support for business decision-making.

2. Data source and cleaning

2.1. Data source

The data comes from the Kaggle website: Coffee Sales。This data records the daily coffee sales amount from March 1, 2024, to March 23, 2025. There are a total of 6 variables and 3636 rows of data. The following are the detailed information of each variable:

Table 2.1 detailed information of variables in sale data

Variable	Data type	Detail
data	Numeric	Format: yyyy-mm-dd
datatime	Character·	Format: mm.s
cash_type	Character	Level 1: card Level 2: cash
card	Character	Card code

money	Numeric	Sale
coffee_name	Character	Level 1: latte Level 2: hot chocolate Level 3: americano Level 4: americano with milk cocoa Level 5: coca Level 6: cortado Level 7:espresso Level 8: cappuccino

2.2. Data cleaning

2.2.1. Convert data format

In this section, we used the “lubridate” package to convert the time variable “date” into a unified standard format, in order to facilitate subsequent time series data forecasting.

2.2.2. Missing value

The missing values in this dataset are represented by a space. To facilitate processing, we convert all missing values to the recognizable form "NA". According to statistics, the variable "card" has 89 missing values. We infer that the missing data is due to transactions made via cash payment not having a card code. These missing values do not affect our analysis, so we simply use "cash" as the value to fill the missing data.

2.2.3. Outliers

As coffee sales data, the sales amount is usually not negative. Therefore, when detecting outliers, we filtered all records with sales amounts less than 0 and found no abnormal situations. In the end, we retained all the original data.

2.2.4. Unify the format of coffee names

To keep the dataset aesthetically pleasing, we used the package "stringr" to standardize the format of the "coffee_name" variable. We cleaned up the whitespace before and after the data fields and changed the character values to title case (capitalizing the first letter of each word).

2.2.5. Convert cash_type and coffee_name into factor

To facilitate visualization and statistical analysis, both the cash_type and coffee_name variables were converted to factor variables. This ensured that these categorical variables were treated appropriately in subsequent analyses.

3. Data analysis and visualization

In this section, we analyzed cardholder spending patterns using Sequence Analysis. We transformed transaction records into time-series "personal behavior chains" to capture the evolution of consumer preferences from early exploration to long-term loyalty.

At the data engineering level, we first applied precise filtering to retain core members with high-frequency repeat purchases. Subsequently, using de-structuring processing and dynamic NA filling techniques, we consolidated the uneven purchase frequencies into a standardized observation matrix.

For behavioral analysis, we adopted a "top-down" observation approach. First, we examined market preferences shifting from dispersion to concentration through aggregate distribution charts. Next, we identified the most popular star products at each stage using mode trajectory charts. Finally, we delved into individual sequences to deconstruct the consumption preferences of core users.

3.1. Retain only card-paying customers with ≥2 purchases

We chose to analyze consumers who use member cards because cash transactions lack the continuity required for sequence data mining. The preprocessing stage focuses on core repeat customers by excluding those with fewer than two purchases, which helps improve the quality of the dataset. Using the member card as an identifier allows us to link multiple transactions together, enabling us to observe how individual preferences evolve over time.

3.2. Create a sequence

To study the evolution of customer consumption behavior, we transformed scattered individual transaction records into "chronologically ordered behavioral sequences":

We grouped members using group_by(card) and employed the summarise() and paste() functions to concatenate all purchase records for each member into complete behavioral chains in temporal order. Subsequently, we employed trimws() to eliminate redundant spaces in product names, ensuring consistency for each coffee_name. Finally, we extracted the raw sequence list representing the customer lifecycle path.

3.3. Convert the sequence to matrix format

To address significant variations in purchase frequency among members, we employed laply() to pad sequences with "NA" values at the end based on the maximum observed sequence length.

This process transforms unstructured lists into a regular observation matrix, ensuring TraMineR can accurately identify customer lifecycles of varying durations.

3.4. Create a sequence object

In order to meet the data input format requirements (matrix format) for sequence trajectory analysis, we implemented the following conversion process:

We used strsplit() to deconstruct the behavior chains, then performed NA padding by dynamically calculating the maximum sequence length.

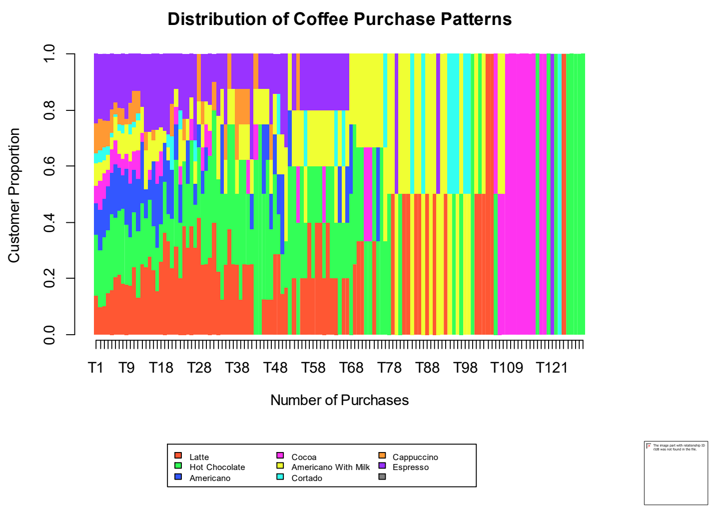
We reconstructed the list object into an observation matrix using do.call(rbind, ...), and extracted all possible 'state sets' (Unique Coffee Types), preparing for the construction of the state transition matrix.

3.5. Sequence Visualization and Behavioral Analysis

We analyze customer behavior from three distinct visual dimensions: overall distribution, modal dominance, and individual trajectories.

3.5.1. State distribution map

seqdplot() reveals how the proportion of each coffee variety in overall consumption evolves with increasing purchase frequency:



Graph3.1 Distribution of Coffee Purchase Patterns

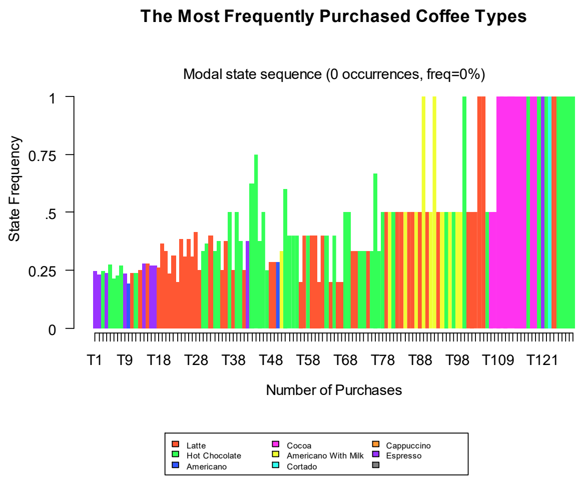
During the initial phase (T1–T23), customers exhibit distinct "experimental" behavior. At this stage, Americano, lattes, and Hot Chocolate hold relatively larger shares, yet the market remains decentralized with balanced distribution across varieties.

By the mid-stage (T31–T71), preferences began shifting toward Latte, Hot Chocolate, and Americano With Milk, with these three categories seeing significantly expanded coverage.

In the late stage (T79–T124), the consumption structure became highly convergent. Cocoa, Latte, and Hot Chocolate formed the brand’s most solid sales foundation and emerged as the primary drivers of high-frequency repeat purchases.

3.5.2. Modal state diagram

seqmsplot() tracks the "best-selling item" (mode state) at each purchase sequence position, with its Y-axis (State Frequency) reflecting the degree of consistency in customer choices:



Graph3.2 The Most Frequently Purchased Coffee Types

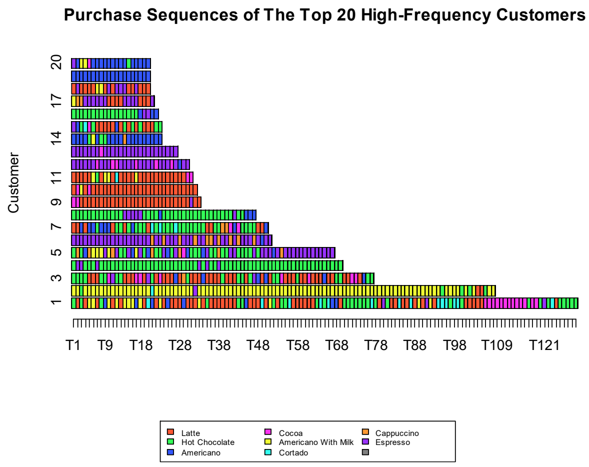
As the horizontal axis (Number of Purchases) moves rightward, bar heights increase significantly. In the early stages (T1-T9), the most frequent variety accounted for only about 25% of purchases. By the T100 phase, frequency surged to 1.0, indicating that long-term customers had identified their preferred coffee.

During the extended period from T18 to T98, Hot Chocolate (green) and Latte (red) frequently alternated as the most popular choice, maintaining a stable frequency above 0.25 to 0.5. This demonstrates that these two products are the "absolute mainstays" sustaining mid-term member activity.

In the later phase (T109 onwards), Cocoa (pink) dominates the chart. This signifies that during the extremely high-frequency consumption phase, long-term consumers exhibit a specific preference shift towards the Cocoa category.

3.5.3. Sequence Index Diagram

We executed seqiplot() to analyze the subtle differences in individual customer purchase sequences.



Graph3.3 Purchase Sequences of The Top 20 High-Frequency Customers

Unlike initial random samples, the Top 20 High-Frequency plot reveals that our most loyal customers are not monolithic.

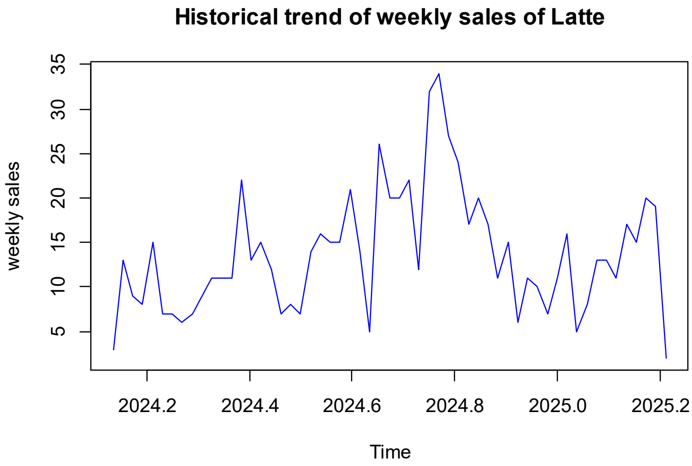
While some (for example Customer 2) exhibit "Single-Product Fixation" on Americano with Milk (Yellow), which serves as a major pillar for the overall sales volume of this category.

Our top contributor (Customer 1) demonstrates a "Multi-Phase Exploration" pattern, eventually stabilizing on Cocoa and Hot Chocolate in the latest stages (T100+). This discovery directly explains why the terminal mode frequencies in Graph 3-2 are largely occupied by pink and green.

4. Mining time series data

4.1. Preprocessing

In this section, we selected all records with "coffee_name" as "Latte" as our research object. The sales data is divided on a weekly basis to study the changing patterns of sales amount and roughly predict future trends. The following is the line chart of the overall data:



Graph 4.1 line plot of weekly sales of latte

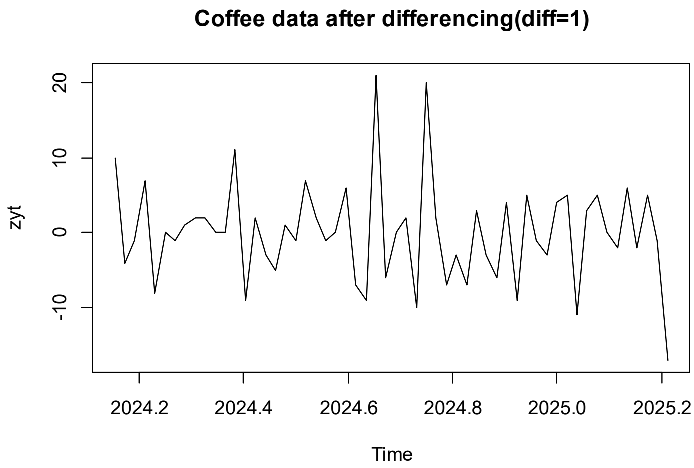
As shown in the figure above, the coffee sales amount fluctuates greatly, but there is no obvious trend. Sales peaked from July to August 2024 but reached a low in February 2025. From this, it is inferred that people have less willingness to spend at the beginning of the year, while their willingness to consume coffee is stronger in the middle of the year. Overall, the data does not meet the premise of stability, and we will further test it with hypothesis testing.

```
        Augmented Dickey-Fuller Test

data:  latte_ts
Dickey-Fuller = -2.2881, Lag order = 3, p-value = 0.4584
alternative hypothesis: stationary
```

Graph 4.2 output of ADF-test

The principle of the Augmented Dickey-Fuller Test is to establish a regression model based on the lagged levels and differences of the time series. By determining whether a unit root exists in the series, it is possible to judge whether the series is stationary (a series with a unit root is non-stationary). As shown in the figure above, when we test the time series data of Latte coffee, we can see that the p-value is 0.4584. This value is much greater than 0.05. This means that there is only about a 55% probability of rejecting the null hypothesis and considering the series to be stationary. This statement is not at all convincing. Therefore, we believe that the series we are studying is non-stationary and needs to be differenced.

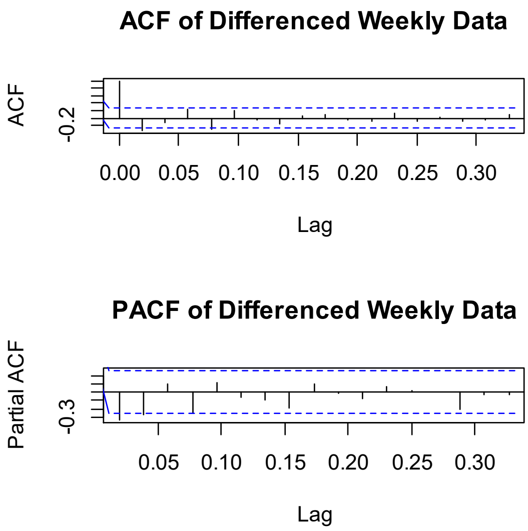


Graph 4.3 plot of differencing data

After the first-order difference, it can be seen that the data tends to be stable, and the p-value in the ADF-test is 0.01, which satisfies the premise of the time series model.

4.2. Arima

Before building the ARIMA model, we first determine the model parameters through ACF and PACF plots.



Graph 4.4 ACF and PACF

Looking at the images, in the ACF plot, lags 1 and 2 are continuously significant; in the PACF plot, lags 1 and 2 are significant. We obtain p and q values of 2. At the same time, we established ARIMA(2,1,2) to select the most suitable model. The following are the details of the models:

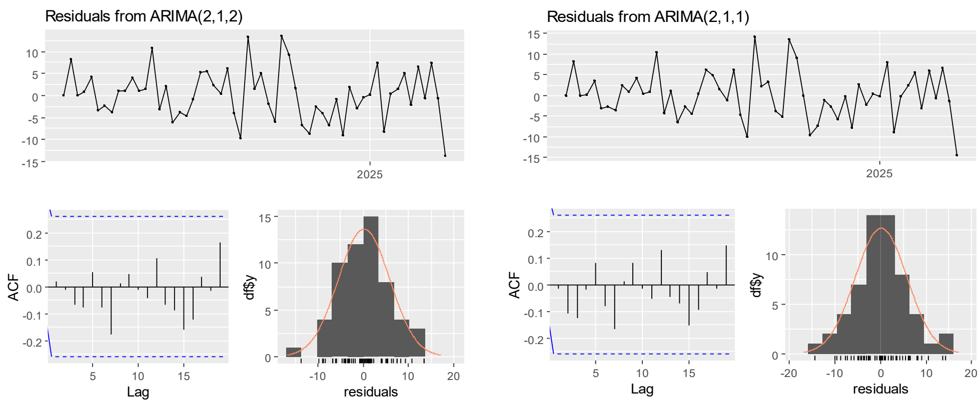
Table 4.1 arima model comparing

Name	Model	AIC value
model_ar2	arima(2,1,2)	364.13
model_ar3	arima(1,0,0)	370.35.

Based on the summary information, the ARIMA(2,1,2) model is suspected of overfitting. This is because the MA(2) term appears redundant, with its coefficient's absolute value being less than its standard error, resulting in a t-statistic less than 2, indicating it primarily fits noise rather than meaningful patterns. Furthermore, the ARIMA(2,1,2) model exhibits poor cost-effectiveness due to its higher number of parameters and a higher AIC compared to simpler models. The additional parameters increase complexity without a corresponding improvement in the fit. Therefore, to address overfitting concerns, we reduce the model order by removing the insignificant MA(2) term, resulting in a simplified ARIMA(2,1,1) model.

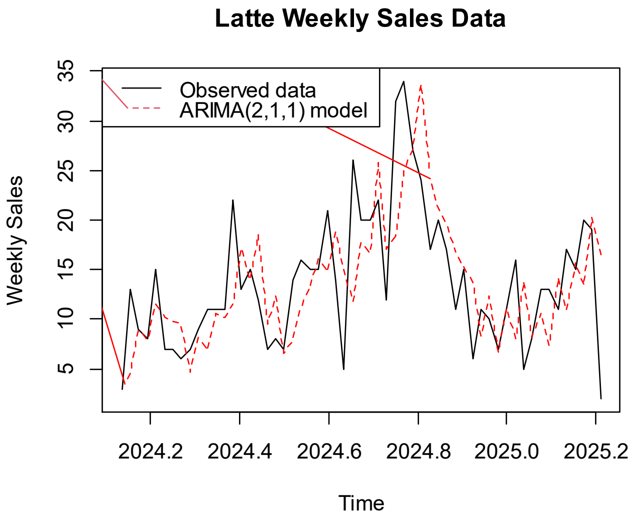
Table 4.2 arima model comparing (2)

Name	Model	AIC value
model_ar2	arima(2,1,2)	364.13
model_ar3	arima(1,0,0)	370.35.
model_ar4	arima(2,1,1)	362.74



Graph 4.5 model diagnostics

After evaluating both the ARIMA(2,1,2) and ARIMA(2,1,1) models, we observed that each produces residuals resembling white noise, indicating a valid model fit. However, the ARIMA(2,1,1) model demonstrates superior performance due to its lower AIC value, suggesting a more parsimonious model without sacrificing accuracy. Furthermore, all parameters within the ARIMA(2,1,1) model exhibit statistical significance, confirming their utility in capturing underlying patterns rather than merely fitting noise. Consequently, we have selected the ARIMA(2,1,1) model as the final choice for its balance of simplicity and effectiveness.

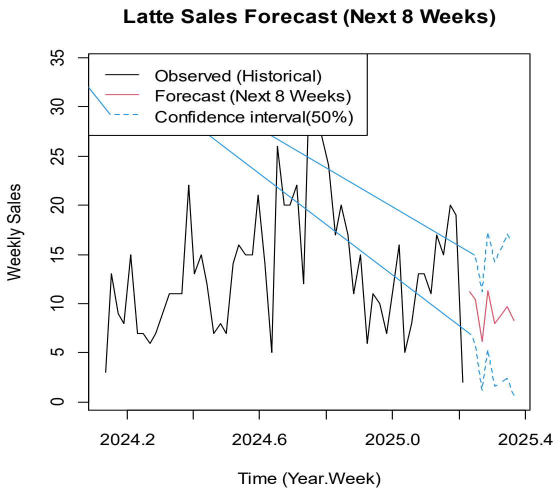


Graph 4.6 model fitting value plot

From this figure, we can see that the fitted values of the model basically match the trend and fluctuations of the true values, indicating that ARIMA(2,1,1) accurately describes the pattern of coffee sales changes. We believe that this model can predict future sales values relatively accurately.

4.3. Forecasting

Based on the existing model, we predicted the sales data changes for the next 8 weeks. The interval confidence level is 50%, resulting in the following figure:



Graph 4.7 foracast next 8 weeks

From the image, we can see that future sales data, although fluctuating, tends to be stable. Since the last data in the dataset has a significant downward trend, which seriously affects the model's judgment, raising the confidence level can lead to counterintuitive results. Based on common sense, we can only ensure that there is a 50% probability that the data will develop within our expected range.

Overall, I think our predictive model has significant room for improvement. As we've seen, the model we used meets all the prerequisites and has relatively good accuracy. We hypothesize that the reason the prediction results are not entirely satisfactory is because the sample data size is too small. Looking at the line chart, the data shows different performance depending on the time of year, generally rising and then falling. If we could obtain more years of coffee sales data, we should be able to better capture the cyclicality of the time series data, and the prediction results would perform better.

5. Summary and advice

5.1. Summary

This study analyzed a coffee brand’s transaction data (March 2024–March 2025) through sequence analysis and time series mining. Sequence analysis revealed that member consumption evolves from early multi-product exploration to mid-term focus on Latte and Hot Chocolate, and finally to late-stage preference for Cocoa among high-frequency users. Time series analysis showed that Latte sales fluctuate with seasonal trends, and the ARIMA(2,1,1) model effectively fits sales patterns, though short-term forecasts are limited by small sample size and lack of long-term cyclical data.

5.2. Advice

5.2.1. Product Strategy: Prioritize promoting lattes, hot chocolate, and cocoa (core products at each stage of consumption); offer combo deals for these products to encourage repeat purchases.

5.2.2. Customer Retention: Provide new members with trial packs of popular products to guide preference formation; offer personalized recommendations based on individual consumption sequences.

5.2.3. Sales Forecasting: Expand data collection to include multiple years of sales data to capture cyclical trends; combine with external factors (such as seasonal promotions) to improve forecast accuracy.

5.2.4. Data Management: Maintain standardized collection of member transaction data to support in-depth analysis of long-term consumption behavior.